PROJECT REPORT ON

TRAVEL AND TOURISM

Submitted in partial fulfilment of the requirement for the award of

BACHELOR OF COMPUTER SCIENCE DEGREE
OF
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Submitted by

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PILATHARA CO-OPERATIVE ARTS AND SCIENCE COLLEGE

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CERTIFICATE

This is to certify that the project "TRAVEL AND TOURISM" submitted during the SIXTH semester BSC COMPUTER SCIENCE as mini project is a result of bonafide work carried out by ANJANA T during the year 2024 – 2025.

Internal Project Guide: External Examiners:

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CERTIFICATE

This is to certify that **ANJANA.T (PL22CCSR19)** student of **PILATHARA CO OPERATIVE ARTS AND SCIENCE COLLEGE** has successfully completed h er academic project entitled **"TRAVEL & TOURISM"** in **Python** under the guidance of ou r senior developers during the p eriod **AUGUST 2024 to FEBRUARY 2025.**

During this period she was found hardworking, punctual & efficient. We wish her a successful future.

For RISS TECHNOLOGIES

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DECLARATION

We hereby declare that the project report entitled “TRAVEL AND TOURISM” Submitted by ANJANA T in fulfilment of requirements for the award of Bachelor of Computer Science is a report of the original work done by us during the period of study in PILATHARA CO-OPERATIVE ARTS AND SCIENCE COLLEGE as centre under the supervision of Mr. ASWIN KRISHNAN V K.

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DATE:

ABSTRACT

The primary objective of this project is to develop a comprehensive system that offers detailed information about destinations and accommodations, and allows users to customize their own travel packages and budgets. This web-based platform will serve as a centralized hub for news, information, and advertisements, reaching millions of potential travelers. The project is especially beneficial for tourists who are unfamiliar with their desired destinations. By leveraging a geographic information system, the platform will provide valuable guidance to tourists and individuals relocating to new cities, ensuring they have access to the best places to visit, stay, and explore. The customizable package and budget feature will enhance the travel experience by empowering users to tailor their trips according to their preferences and financial constraints, delivering reliable and relevant information to users.

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INTRODUCTION

1.1 INTRODUCTION

The **TRAVEL AND TOURISM** is a comprehensive platform designed to simplify the travel booking and management process for users, travel agencies, and administrators. It consists of three primary modules: **ADMIN**, **TRAVEL AGENCY**, and **USER**.

The **ADMIN** module allows administrators to manage users, travel agencies, and hotels, as well as configure system settings. The **TRAVEL AND TOURISM** module enables agencies to list, manage, and update travel packages, handle bookings, and interact with customers. The **USER** module provides customers with the ability to browse and book travel packages, customize their own package with budget, make payments, and manage their bookings, while also offering feedback through ratings and reviews. The project aims to create an efficient and seamless experience for all users involved in the travel process.

The project is done using PYTHON(DJANGO), MYSQL and also the user module is mainly done using ANDROID.

1.2 OBJECTIVE

The **Travel and Tourism Project** aims to streamline travel booking and management by offering a comprehensive platform with three core modules: Admin, Travel Agency, and User. The primary objectives include simplifying travel package browsing and booking for users, enabling travel agencies to efficiently manage and update packages, and providing administrators with control over user management, hotels, and system settings. Additionally, the platform facilitates customized package, secure payments, user feedback, and communication, ensuring a seamless and efficient experience for all stakeholders in the travel process.

2.SYSTEM STUDY

2.1 EXISTING SYSTEM

In the present system a customer has to approach various agencies to find details of places and to book packages. This often requires a lot of time and effort. A customer may not get the desired information from these offices and often the customer may be misguided. It is tedious for a customer to plan a particular journey and have it executed properly.

2.1.1 DRAWBACKS OF EXISTING SYSTEM

- ✓ Due to the heavy traffic networks the process gets delayed.
- ✓ It is available in many cities and towns but not in all. It needs further expansion.
- ✓ Difficult to use these as they are not that user-friendly.

2.2 PROPOSED SYSTEM

The proposed travel and tourism system aims to enhance the booking experience by incorporating advanced security features and a user-friendly interface available on both web and Android platforms. The system will allow users to register and log in to the platform, providing a secure and personalized experience. Travel agencies will have control over their listings, including approving or rejecting bookings, updating real-time availability and pricing, and handling customer interactions and reviews.

Users can search for and compare travel services from multiple agencies, with the added benefit of customizing their own travel packages and budgets. The platform will utilize AI-driven recommendations to suggest additional services or upgrades tailored to the user's preferences. To finalize a booking, users will enter a secure process to confirm the transaction, ensuring reliable and safe bookings.

The system will support real-time availability, secure payment gateways, and personalized user profiles. Users will have access to their booking history, be able to leave reviews, and communicate with travel agencies for support and inquiries. By integrating these features, the proposed system aims to provide a seamless, secure, and personalized travel planning experience for users across both web and mobile interfaces.

2.2.1 ADVANTAGES

- ✓ Enhanced travel efficiency
- ✓ Time saving
- ✓ Cost efficiency
- ✓ Flexibility and Personalization

3.SYSTEM ANALYSIS

System analysis is the detailed study of the various operations performed by the system and relationship within and outside the system. It can also be defined as the way of studying a system with an eye on solving it problem using computers. It is the most essential part of the development of a project. During system analysis data are collected on the available files, decision points and transactions handled by the present system. System analysis must carry out a customary approach to the use of computers for the problem solving.

System analysis is concerned with studying the existing system identifying the problems and creation of the requirement specification document. In the system analysis I study about the system to be computerized and the entire problem with the present system are pinpointed. System Analysis is conducted with the following objectives

- Identify the customer's need
- Evaluate the system concept for feasibility
- Perform economic analysis
- Technical analysis
- Allocate function to hardware, software, people, database and other system elements
- Establish cost and schedule constraints
- Create system definition that forms the foundation for all subsequent engineering works.

The system analysis was planned and conducted in three stages, involving an initial investigation, feasibility and detailed analysis.

Once upon a time software development consisted of a programmer writing code to solve a problem or automate a procedure. Nowadays, systems are so big and complex that teams of architects, analysts, programmers, testers and users must work together to create the millions of lines of custom-return code that drive our enterprises.

Assuming that a new system is to be developed, the next phase is system analysis.

Analysis involved a detailed study of the current system, leading to specifications of a new system.

The main points to be discussed in system analysis are:

- Specification of what the new system is to accomplish based on the user requirements.
- Functional hierarchy showing the functions to be performed by the new system and their relationship with each other.

3.1 FEASIBILITY STUDY

Feasibility study is a procedure that identifies, describes and evaluates candidate system and selects the best system for the job. An estimate is made of whether the identified user needs may be satisfied using current software and hardware technologies. The study will decide if the proposed system will be cost effective from a business point of view and if it can be developed given existing budgetary constraints. The key considerations involved in the feasibility analysis are

- ❖ Economic Feasibility
- ❖ Technical Feasibility
- ❖ Social or behavioural Feasibility

❖ <u>Economic Feasibility</u>

Economic feasibility procedure involves the analysis of cost involved to set up the new system. The cost of conducting the investigation and development of the system were not high and affordable. The system reduces the cost, effort and time of projects that has been carried out in organizations. The system was found to be economically feasible.

Economic analysis is the most frequently used technique for evaluating the effectiveness of a proposed system. More commonly known as cost/benefit analysis the

❖ Technical Feasibility

This concerned with specifying equipment and software that will successfully satisfy the user requirement. In examining the technical feasibility, configuration of the system is given more importance than the actual hardware. Out of all types of feasibility generally is the most difficult to determine.

It does not need heavy investments for the enhancements. It can run on any compatible device and is thus technically feasible in all environments.

❖ Social or Behavioural Feasibility

Each of the alternative technical solutions that emerge must be evaluated for its social implications. The need of various people affected by the proposed system must be taken into account. People are inherently resistant to changes and computers have been supposed to facilitate the change.

An estimate should be of how strong the reaction of the people is likely to have towards the development of a computerized system. It is the common knowledge that the installation of the new software has something to do with the turn over, transfers retraining and changes in job status and so. It is understandable that the introduction of a system requires special effort to educate and train the persons on new ways of conducting tests.

4.SYSTEM SPECIFICATIONS

4.1 SYSTEM REQUIREMENT

Actual user community participation and their requirements analysis is the key to success of any new information system. To carry out this work, identification of users who will actually use the system is foremost. Users at every level were even given an opportunity to define their goals, objectives and their respective information needs.

The system requirement study involves the study of existing system's inputs, outputs and costs. The tools used in data collection are:

- Review of written documents
- On-site observations
- Interviews and Questionnaires.

✓ The findings of users and other related exercise to access particular user needs are summarized below concisely:

- The system must provide a graphical user interface.
- The system must be user friendly
- Redundancy must reduce at the maximum level.
- Discrepancies should be avoided.
- Security systems must be provided.

4.2 HARDWARE REQUIREMENTS

It is recommended that for optimal performance, the following minimum hardware are installed on the server on which the portal is hosted, as well as on clients that access the portal.

- INPUT DEVICE : MOUSE,KEYBOARD
- OUTPUT DEVICE : MONITOR
- MEMORY : 4 GB RAM(MINIMUM)
- PROCESSOR : Intel core i3 or above

4.3 SOFTWARE REQUIREMENTS

The software requirement specification (SRS) and hardware specification forms the basis of software development. The purpose of software requirement specification is to improve understandability and needs of the software. This provides a validation of the final product and its future development.

For the proposed system to work properly, it is necessary that following software are installed and running on the server / client.

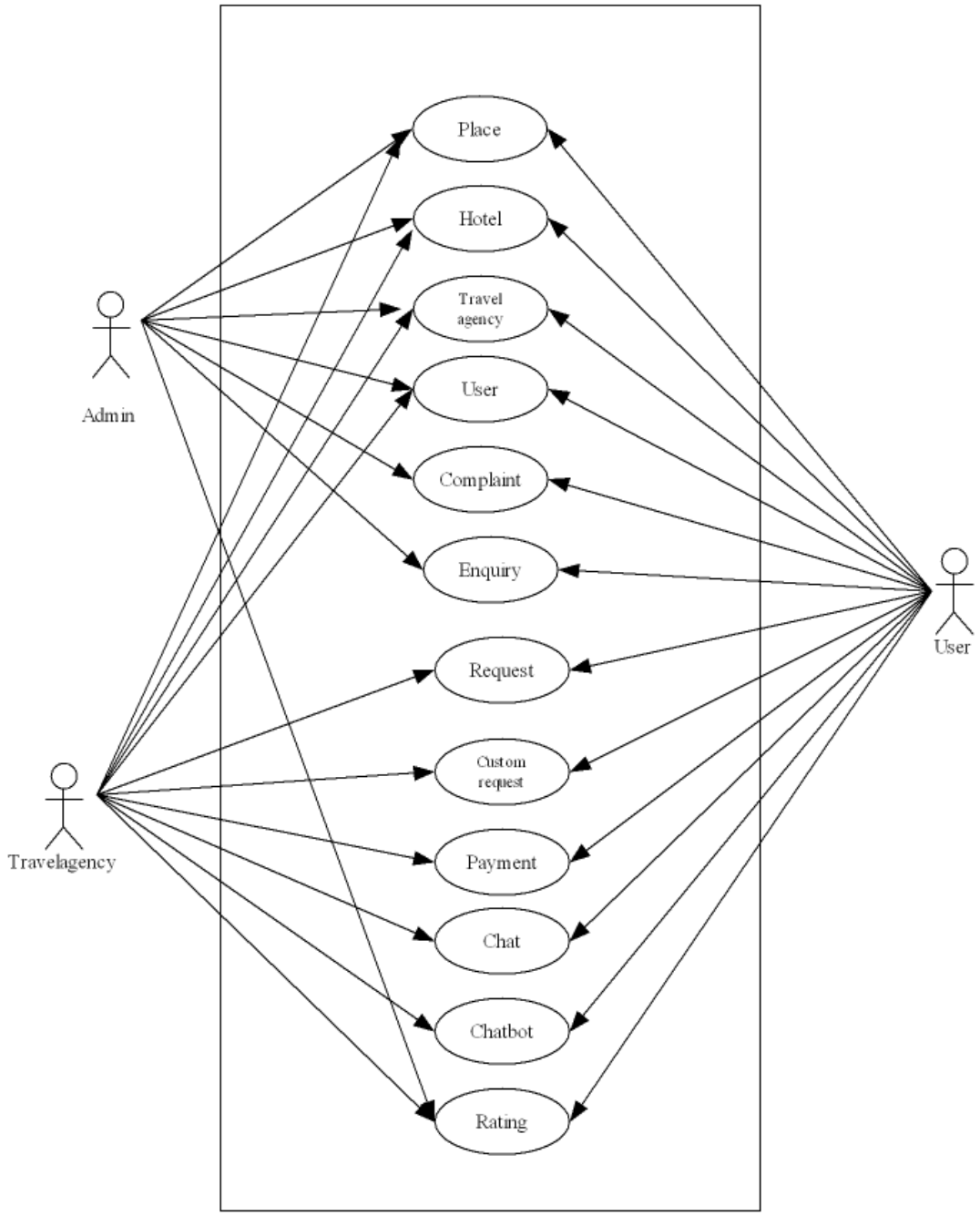
- OPERATING SYSTEM : WINDOWS 8 for better performance
- FRONT END : HTML, JS, CSS, XML
- BACK END : MySQL
- LANGUAGE : Python
- IDE USED : Android Studio, PyCharm

4.4 IDENTIFICATION OF USE-CASES

A use-cases represents the functionality of an actor. It is defined as a set of actions performed by a system, which yields an observable result. An ellipse containing its name inside the ellipse or below it represents it. It is placed inside the system boundary and connected to an actor with an association. This shows how the use cases and the actor interact. To find out the use cases, ask the following questions to each of the actors.

- Which functions does the actor require from the system? What does the actor need to do?
- Does the actor need to read, create, destroy, modify or store some kind of information in the system?
- Does the actor has to be notified about events in the system, or does the actor need to notify the system about something? What do those the events represent in terms of functionality?
- Could the actor's daily work be simplified or made more efficient by adding new functions to the system (typically functions which are currently not automated in the system)

4.5 USECASE DIAGRAM



5.SOFTWARE DESCRIPTION

PYTHON

Python is used to create algorithms — or sets of instructions — that can be read and implemented by a computer. It is one of countless programming languages used to develop software on the web, and in apps… perhaps including the app you're using to read this blog right now! Other programming languages you might have heard of include Java script, Ruby, C++, and PHP.

Guido Van Rossum, first released the language in 1991 (fun fact: he named it after Monty Python!) Python is now on its third major release, and has enjoyed a recent surge in popularity. Its syntax (the set of rules that govern its structure) focuses on readability and ease of use, making it a great programming language for beginners. That's also why it's perfect for rapid prototyping.

Python is what is known as a general-purpose programming language. Simply, this means it has a wide variety of applications. For instance, it is commonly used for web and software development. However, it is also popular for backend(for server side), data analytics, and for producing system scripts.

Python is a high-level programming language. This means it comes with many operations built-in behind the source code. Using a process known as 'abstraction', programmers can hide the majority of what the computer needs to do, making it easier for coders to focus on achieving their desired goal. In contrast, a low-level language requires much more consideration for how a computer h.

To make this clearer, imagine you have a list of names that you want to sort in alphabetical order. In a high-level language, you'd do this by executing a command such as: 'sort alphabetically'. In a low-level language, you would have to be far more specific, e.g. 'A comes before B, B comes before C', and so on.

While the strengths and limitations of high- and low-level programming are quite nuanced, in general, high-level languages are preferred. They minimize complexity and are more efficient — they're also easier for beginners to learn.

- **Python is general purpose:** It supports a number of programming paradigms, meaning it can be used for many different types of project.
- **The code is easy to learn:** Python's syntax is intuitive and easy to grasp, making it a great language for beginners and experienced programmers alike.
- **It is platform-independent:** Python's code can be run on any computer operating system, with only minor tweaks required.
- **It is suitable for fast coding:** Python allows for quick coding and easy debugging (fixing errors) making it perfect for rapid application development.
- **Strong standard library:** Python's standard library is well-stocked. It has a wide range of modules, syntax guides, and other supporting documentation.
- **Massive range of third-party libraries:** Python has a huge selection of tools, reusable code, and third-party libraries that cover a wide range of applications.
- **It has a big online community:** Python's active community is constantly evolving, problem-solving, and sharing its open-source code.

➢ System scripts and automation

Python was initially developed as a back-end scripting language, used to automate 'mundane' behind-the-scenes tasks. While scripts have grown more sophisticated over the years, Python remains popular. It can be used to write scripts in a fraction of the time it would take to do so using a more complex language. Today, Python is used to automate everything from web searches and content downloads, to completing online forms, and sending emails.

➢ Web and app development

Python is also a popular language among web developers. Google, Instagram, and reddit are all examples of sites built using Python. Thanks to Django and Flask, two extensive Python web frameworks (which are foundation platforms on which software can be built), Python

web development is a relatively straightforward process. Django is one of the most well-known web frameworks in existence, providing a robust architecture of free reusable code.

- **Scientific computing and data analytics**

Astronomers, biologists, and neuroscientists all use Python. Even organizations like CERN and NASA…pretty cool, Better yet, thanks to an enthusiastic and active community, Python boasts an almost endless pool of third-party libraries. These include general data manipulation tools, such as Panda, to niche frameworks like PsychoPy (used within experimental psychology). Python is also easily scalable—on the most basic level, it can output data in a simple .csv file. However, it can also produce formats suited to ingestion by a variety of more complex systems.

- **Artificial intelligence (AI) and machine learning**

Within machine learning, Python is used for everything from speech and image recognition, to the algorithms used by companies like Netflix. It is even used within the financial sector, most notably as the architecture behind the New York Stock Exchange. This is testament to its reliability and security. There are a huge range of Python machine learning libraries available, such as Tensor flow—a high-level neural network library. Python's uses in the area of AI are only going to grow.

- **Gaming and special effects**

Developers have been using Python to create graphic user interfaces (GUI) for quite some time. However, with a range of Python 3D animation packages available, it has also found applications in computer gaming and special effects. Libraries available for budding game developers include Pygame, Pysoy and Pykyra. The digital special effects house ILM (whose work has appeared in all the recent Star Wars and Marvel films) also uses Python.

MYSQL

The MySQL database has become the world's most popular open source database because of its consistent fast performance, high reliability and ease of use. It's used on every

continent by individual Web developers as well as many of the world's largest and fastest-growing organizations to save time and money powering their high-volume Websites, business-critical systems and packaged software -- including industry leaders such as Yahoo!, Alcatel-Lucent, Google, Nokia, YouTube, and Zappos.com.

Not only is MySQL the world's most popular open-source database, it's also become the database of choice for a new generation of applications built on the LAMP stack (Linux, Apache, MySQL, PHP / Perl / Python.) MySQL runs on more than 20 platforms including Linux, Windows, OS/X, HP-UX, AIX, Netware, giving one the kind of flexibility that puts him in control.

MySQL, the most popular Open-Source SQL database management system, is developed, distributed, and supported by MySQL AB. MySQL AB is a commercial company, founded by the MySQL developers. It is a second-generation Open Source company that unites Open-Source values and methodology with a successful business model.

MySQL is a database management system.

A database is a structured collection of data. It may be anything from a simple shopping list to a picture gallery or the vast amounts of information in a corporate network. To add, access, and process data stored in a computer database, one needs a database management system such as MySQL Server. Since computers are very good at handling large amounts of data, database management systems play a central role in computing, as standalone utilities, or as parts of other applications.

MySQL is a relational database management system.

A relational database stores data in separate tables rather than putting all the data in one big storeroom. This adds speed and flexibility. The SQL part of “MySQL” stands for “Structured Query Language.” SQL is the most common standardized language used to access databases and is defined by the ANSI/ISO SQL Standard. The SQL standard has been evolving since 1986 and several versions exist.

MySQL software is Open Source.

Open-Source means that it is possible for anyone to use and modify the software. Anybody can download the MySQL software from the Internet and use it without paying anything.

The MySQL Database Server is very fast, reliable, and easy to use.

MySQL Server also has a practical set of features developed in close cooperation with its users. MySQL Server was originally developed to handle large databases much faster than existing solutions and has been successfully used in highly demanding production environments for several years. Although under constant development, MySQL Server today offers a rich and useful set of functions. Its connectivity, speed, and security make MySQL Server highly suited for accessing databases on the Internet.

MySQL Server works in client/server or embedded systems.

The MySQL Database Software is a client/server system that consists of a multithreaded SQL server that supports different back-ends, several different client programs and libraries, administrative tools, and a wide range of application programming interfaces (APIs). MySQL Server also comes as an embedded multi-threaded library that one can link into his application to get a smaller, faster, easier-to-manage standalone product.

APACHE HTTP SERVER

To run any dynamic web based application one needs a web-server. Web-server is a software designed to capture user's request sent from a web client (like a web browser), process it and serve the requested resource to the client.

Apache is an Open-Source web-server and is one of the most commonly used one. News Portal, being an online news system, can be used by huge number of users simultaneously. So the system needs to take the bigger view at its users. Apache servers come cheap and so the system had to be designed for it.

Apache Software exists to provide robust and commercial-grade reference implementations of many types of software. It must remain a platform upon which individuals and institutions can build reliable systems, both for experimental purposes and for mission-critical purposes. Furthermore, the Apache Software Foundation is an organic entity; those who benefit from this software by using it, often contribute back to it by providing feature enhancements, bug fixes, and support for others in public lists and newsgroups. The effort expended by any particular individual is usually fairly light, but the resulting product is made very strong.

HTML

HTML - HyperText Markup Language is the predominant markup language for web pages. It provides a means to describe the structure of text-based information in a document by denoting certain text as links, headings, paragraphs, lists, and so on and to supplement that text with interactive forms, embedded images, and other objects.HTML is written in the form of tags, surrounded by angle brackets. HTML can also describe, to some degree, the appearance and semantics of a document, and can include embedded scripting language code (such as JavaScript) which can affect the behavior of Web browsers and other HTML processors.

HTML is also often used to refer to content in specific languages, such as a MIME type text/HTML, or even more broadly as a generic term for HTML, whether in its XML descended form (such as XHTML 1.0 and later) or its form descended directly from SGML (such as HTML 4.01 and earlier). By convention, HTML format data files use a file extension .HTML or .html.

HTML markup consists of several key components, including elements (and their attributes), character-based data types, and character references and entity references. Another important component is the document type declaration. Elements are the basic structure for HTML markup. Elements have two basic properties: attributes and content. Each attribute and each element's content has certain restrictions that must be followed for an HTML document to be considered valid. An element usually has a start tag (e.g.:<element-name>) and an end tag (e.g. </element-name>). The element's attributes are contained in the start tag and content

is located between the tags (e.g.:<element-name attribute="value">Content</element-name>). Some elements, such as
, do not have any content and must not have a closing tag. Listed below are several types of markup elements used in HTML.

CSS

Style sheets describe how documents are presented on screens, in print, or perhaps how they are pronounced. W3C has actively promoted the use of style sheets on the Web since the Consortium was founded in 1994. The Style Activity has produced several W3C Recommendations (CSS1, CSS2, XPath, XSLT). CSS especially is widely implemented in browsers.

By attaching style sheets to structured documents on the Web (e.g. HTML), authors and readers can influence the presentation of documents without sacrificing device-independence or adding new HTML tags.

The easiest way to start experimenting with style sheets is to find a browser that supports CSS. The W3C Style Activity is also developing XSL, which consists of a combination of XSLT and "Formatting Objects" (XSL-FO).

The unique features are that CSS can be used to style HTML & XML documents. XSL, on the other hand, is able to transform documents. For example,

XSL can be used to transform XML data into HTML/CSS documents on the Web server. This way, the two languages complement each other and can be used together. CSS and XSL use the same underlying formatting model and designers therefore have access to the same formatting features in both languages. W3C will work hard to ensure that interoperable implementations of the formatting model are available.

JAVASCRIPT

JavaScript is a scripting language most often used for client-side web development. It was the originating dialect of the ECMAScript standard. It is a dynamic, weakly typed, prototype-based language with first-class functions. JavaScript was influenced by many languages and was designed to look like Java, but be easier for nonprogrammers to work

with.

Although best known for its use in websites (as client-side JavaScript), JavaScript is also used to enable scripting access to objects embedded in other applications.

JavaScript, despite the name, is essentially unrelated to the Java programming language, although both have the common C syntax, and JavaScript copies many Java names and naming conventions. "JavaScript" is a trademark of Sun Microsystems. It was used under license for technology invented and implemented by Netscape Communications and current entities such as the Mozilla Foundation.

JavaScript is Structured Programming

JavaScript supports all the structured programming syntax in C (e.g., if statements, while loops, switch statements, etc.). One partial exception is scoping: C-style block-level scoping is not supported. However, JavaScript 1.7 supports block-level scoping with the let keyword.

JSON

JSON stands for JavaScript Object Notation, which is a lightweight, text-based, and language-independent data interchange format123. It is based on a subset of the JavaScript programming language and uses a simple syntax to represent data structures, including objects, arrays, strings, numbers, and boolean values45. JSON is commonly used for transmitting data between a server and a web application, or between different parts of a web application5. It is also used for storing and exchanging data in various programming languages and platforms, including JavaScript, Python, Java, and .NET

.

ANDROID

Android is a mobile operating system developed by Google. It is based on the Linux kernel and is designed primarily for touchscreen mobile devices such as smartphones and tablets. Android was first released in 2008 and has since become the most widely used mobile operating system in the world. Android has a number of features that make it a popular choice for mobile device manufacturers and app developers. These include:

1. Open Source: Android is an open source operating system, which means that its source code is freely available to developers. This has led to the development of a large and active community of developers who create apps, custom ROMs, and other

modifications to the operating system. 2. Customizable User Interface: Android allows for a high degree of customization of the user interface. This means that device manufacturers can create their own unique user interface designs that differentiate their products from those of competitors. 3. Google Play Store: The Google Play Store is the primary source of apps for Android devices. It has a vast library of apps, including both free and paid apps, and is accessible from any Android device. 4. Multitasking: Android allows for multitasking, which means that users can run multiple apps simultaneously. This is particularly useful for productivity apps and social media, where users often need to switch between multiple apps. 5. Integration with Google Services: Android integrates with a range of Google services, including Google Search, Google Maps, Google Drive, and Google Assistant. This provides users with access to a wide range of useful features and services. Overall, Android is a powerful and flexible operating system that is designed to meet the needs of a wide range of users and device manufacturers. Its open source nature, customization options, and integration with Google services make it a popular choice for mobile device manufacturers and app developers alike.

XML

XML (Extensible Markup Language) is a markup language similar to HTML, but without predefined tags to use. Instead, you define your own tags designed specifically for your needs. This is a powerful way to store data in a format that can be stored, searched, and shared. Most importantly, since the fundamental format of XML is standardized, if you share or transmit XML across systems or platforms, either locally or over the internet, the recipient can still parse the data due to the standardized XML syntax.

6.PROJECT DESCRIPTION

6.1 PROBLEM DEFINITION

Develop an India-focused travel booking platform named **TRAVELA** that allows users to compare and book services from multiple travel agencies, featuring real-time availability and secure payments. Enhance the user experience with personalized recommendations and AI-driven suggestions.

6.2 OVERVIEW OF THE PROJECT

The project aims to develop a web-based and Android platform for travel and tourism bookings, providing a user-friendly interface for both travelers and travel agencies.

The system will support functionalities such as registration, login, and user profile management for travelers and agency representatives.

Key features include search and booking capabilities, real-time availability, secure payments, and user reviews. Additionally, the platform will facilitate communication between travelers and travel agencies, ensuring a seamless and personalized booking experience across web and mobile devices.

6.3 MODULE DESCRIPTION

The project aims to develop a comprehensive web-based and Android platform tailored for the travel and tourism industry. The administrator has the ultimate control over the entire portal, with the ability to manage travel agencies, view user feedback, and oversee registrations. This platform will provide a user-friendly interface for both travelers and travel agencies, supporting functionalities such as registration, login, and user profile management. Travel agencies will have the ability to manage their listings, including approving or rejecting bookings, updating real-time availability and pricing, and handling customer interactions and reviews.

Users will be able to search for and book travel services from multiple agencies through both web and Android interfaces. They can view their booking history, manage their profiles, and communicate with travel agencies for support and enquiries. Additional features include personalized recommendations, secure payment integration, and notifications about bookings, promotions, and updates. This will ensure a seamless and enjoyable booking experience for travelers across platforms.

The system will implement various security measures to protect users' data and transactions. These measures include user authentication through email and mobile verification, encryption of sensitive information, and real-time monitoring of suspicious activities. By providing a secure and efficient platform, the project aims to enhance the travel planning process and create a reliable solution for the travel and tourism industry.

Modules

1. ADMIN Module:

    - Login
    - Place Management
    - Hotel Management
    - View Registered Travel Agency And Approve
    - View Approved Travel Agency
    - View Complaints and Send Reply
    - View Enquiry
    - View App Rating

2. TRAVEL AGENCY Module:

    - Registration
    - Login
    - Food Menu Management
    - Package Management
    - View Package Request And Verify
    - Visiting Place Management
    - Chat
    - View Custom Request And Add Amount
    - View Enquiry And Reply
    - View Rating
    - View Approved Package Request And Add Guide
    - View User Approved Custom Request And Add Guide
    - View Package Payment Status
    - View Custom Payment Status

3. USER module:

- Registration
- Login
- View Profile and Update
- Add Custom package (with places) Management
- View budget and approve
- View approved custom package
- View package and send request
- View package status
- Enquiry management
- Send Rating
- View hotel
- Chat with travel agency
- Send complaint and view reply
- Change password
- Pay for custom package
- Pay for package
- Wishlist
- Chat bot
- Search

6.4 DATABASE

The project uses the following tables:

1. Login
2. Place
3. Hotel
4. Travelagency
5. User
6. Complaint
7. Enquiry
8. Foodmenu
9. Package
10. Package_food
11. Visitingplace
12. Requests
13. Chat
14. Rating
15. Custompackage
16. Budget
17. Wishlist
18. Apprating

Table6.4.1- Login

Field Name	Type(size)	Constraints	Description
id	Int (20)	Primary key	id
username	Varchar (200)	NOT NULL	Name
password	Varchar (200)	NOT NULL	password
usertype	Varchar (200)	NOT NULL	type

Table 6.4.2 – Place

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
placename	Varchar(20)	NOT NULL	name
image	Varchar(200)	NOT NULL	Image of place
description	Varchar(200)	NOT NULL	Description of image
state	Varchar(200)	NOT NULL	State
district	Varchar(200)	NOT NULL	District

Table 6.4.4 – hotel

Field Name	Type(size)	Constraints	Description
id	Int(50)	Primary key	id
hotelname	Varchar(200)	NOT NULL	Name of hotel
image	Varchar(200)	NOT NULL	Image of hotel
price	Varchar(200)	NOT NULL	Price
facilities	Varchar(200)	NOT NULL	Facilities
latitude	Varchar(200)	NOT NULL	Latitude
longitude	Varchar(200)	NOT NULL	Longitude
contact	Varchar(200)	NOT NULL	Contact
email	Varchar(200)	NOT NULL	Email

Table 6.4.5 – travelagency

Field Name	Type(size)	Constraints	Description
id	Int(20)	Primary key	id
agencyname	Varchar(200)	NOT NULL	name
email	Varchar(200)	NOT NULL	Email
contact	Varchar(200)	NOT NULL	Contact
latitude	Varchar(200)	NOT NULL	Latitude
longitude	Varchar(200)	NOT NULL	Longitude
images	Varchar(200)	NOT NULL	Image of travelagency
description	Varchar(200)	NOT NULL	Description
LOGIN	int	NOT NULL	Login id

Table 6.4.6 – user

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
name	Varchar(200)	NOT NULL	username
email	Varchar(200)	NOT NULL	email
phoneno	Varchar(200)	NOT NULL	number
place	Varchar(200)	NOT NULL	Place of user
post	Varchar(200)	NOT NULL	Post of user
pincode	Varchar(200)	NOT NULL	Pincode of user
latitude	Varchar(200)	NOT NULL	Latitude
longitude	Varchar(200)	NOT NULL	Longitude
LOGIN	int	NOT NULL	Login id

Table 6.4.7- complait

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
complaint	Varchar(200)	NOT NULL	Complaint
complaintdate	Varchar(200)	NOT NULL	Date
reply	Varchar(200)	NOT NULL	Reply
replydate	Varchar(200)	NOT NULL	Reply date
USER	int	NOT NULL	User id

Table 6.4.8- enquiry

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	Id
enquiry	Varchar(200)	NOT NULL	Enquiry
enquirydate	Varchar(200)	NOT NULL	Date
reply	Varchar(200)	NOT NULL	Reply
replydate	Varchar(200)	NOT NULL	Reply date
USER	int	NOT NULL	User id
TRAVELAGENCY	int	NOT NULL	Travelagency id

Table 6.4.9- foodmenu

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	Id
item	Varchar(200)	NOT NULL	Item
cost	Varchar(200)	NOT NULL	Cost
description	Varchar(200)	NOT NULL	Description
image	Varchar(200)	NOT NULL	Image
TRAVELAGENCY	int	NOT NULL	Travelagency id

Table 6.4.10- package

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
package	Varchar(200)	NOT NULL	package
cost	Varchar(200)	NOT NULL	Cost of package
numberofpeople	Varchar(200)	NOT NULL	Number of people
days	Varchar(200)	NOT NULL	days
TRAVELAGENCY	int	NOT NULL	Travelagency id

Table 6.4.11- package_food

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
type	Varchar(200)	NOT NULL	Type
PACKAGE	Varchar(200)	NOT NULL	Package id
FOOD	Varchar(200)	NOT NULL	Food id

Table 6.4.12- visitingplace

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
day	Varchar(200)	NOT NULL	day
PLACE	Varchar(200)	NOT NULL	Place id
PACKAGE	Varchar(200)	NOT NULL	Package id

Table 6.4.7- requests

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
status	Varchar(200)	NOT NULL	Status of request
date	Varchar(200)	NOT NULL	Date od request
payment	Varchar(200)	NOT NULL	Payment
paymentmode	Varchar(200)	NOT NULL	Payment mode
paymentdate	Varchar(200)	NOT NULL	payment date
guide	Varchar(200)	NOT NULL	Guide
guidecontact	Varchar(200)	NOT NULL	Guide contact
PACKAGE	int	NOT NULL	Package id
USER	int	NOT NULL	User id

Table 6.4.7- chat

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
message	Varchar(200)	NOT NULL	message
type	Varchar(200)	NOT NULL	type
date	Varchar(200)	NOT NULL	Date of chat
TRAVELAGENCY	int	NOT NULL	Travelagency id
USER	int	NOT NULL	User id

Table 6.4.7- rating

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
rating	Varchar(200)	NOT NULL	rating
review	Varchar(200)	NOT NULL	review
date	Varchar(200)	NOT NULL	Date of rating
TRAVELAGENCY	int	NOT NULL	Travelagency id
USER	int	NOT NULL	User id

Table 6.4.7- custompackage

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
date	Varchar(200)	NOT NULL	Date of custompackage
startdate	Varchar(200)	NOT NULL	Starting date
expectedamount	Varchar(200)	NOT NULL	Expected amount of custompackage
place	Varchar(200)	NOT NULL	place
noofdays	Varchar(200)	NOT NULL	Number of days
noofpeople	Varchar(200)	NOT NULL	Number of people
payment	Varchar(200)	NOT NULL	Payment
paymentmode	Varchar(200)	NOT NULL	Mode of payment
paymentdate	Varchar(200)	NOT NULL	Payment date
paymentstatus	Varchar(200)	NOT NULL	Payment status
USER	int	NOT NULL	User id

Table 6.4.7- budget

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
amount	Varchar(200)	NOT NULL	amount
notee	Varchar(200)	NOT NULL	note
status	Varchar(200)	NOT NULL	status
guide	Varchar(200)	NOT NULL	Guide
guidecontact	Varchar(200)	NOT NULL	Guide contact
TRAVELAGENCY	int	NOT NULL	Travelagency id
CUSTOMPACKAGE	int	NOT NULL	Custompackage id

Table 6.4.7- wishlist

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
USER	int	NOT NULL	User id
PACKAGE	int	NOT NULL	Package id

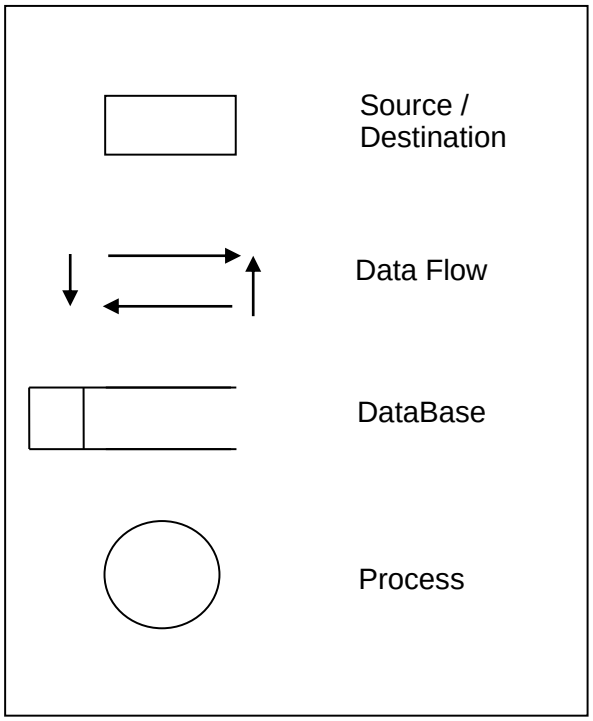
Table 6.4.7- apprating

Field Name	Type(size)	Constraints	Description
id	Int(20)	NOT NULL	id
rating	Varchar(200)	NOT NULL	Rating
review	Varchar(200)	NOT NULL	Review
date	Varchar(200)	NOT NULL	Date of app rating
USER	int	NOT NULL	User id

6.5 DATA FLOW DIAGRAMS (DFD)

Data Flow Diagram is used to represent input to, and output from processes, as well as data written into storage, or retrieved from storage.

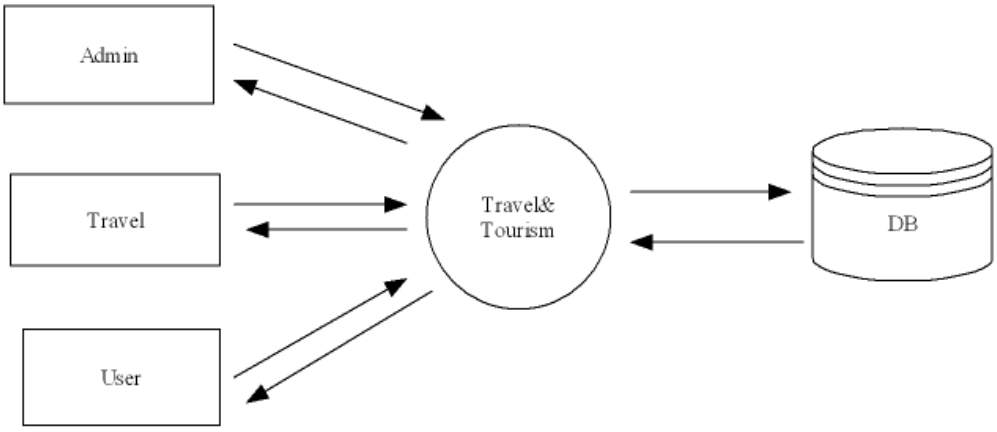
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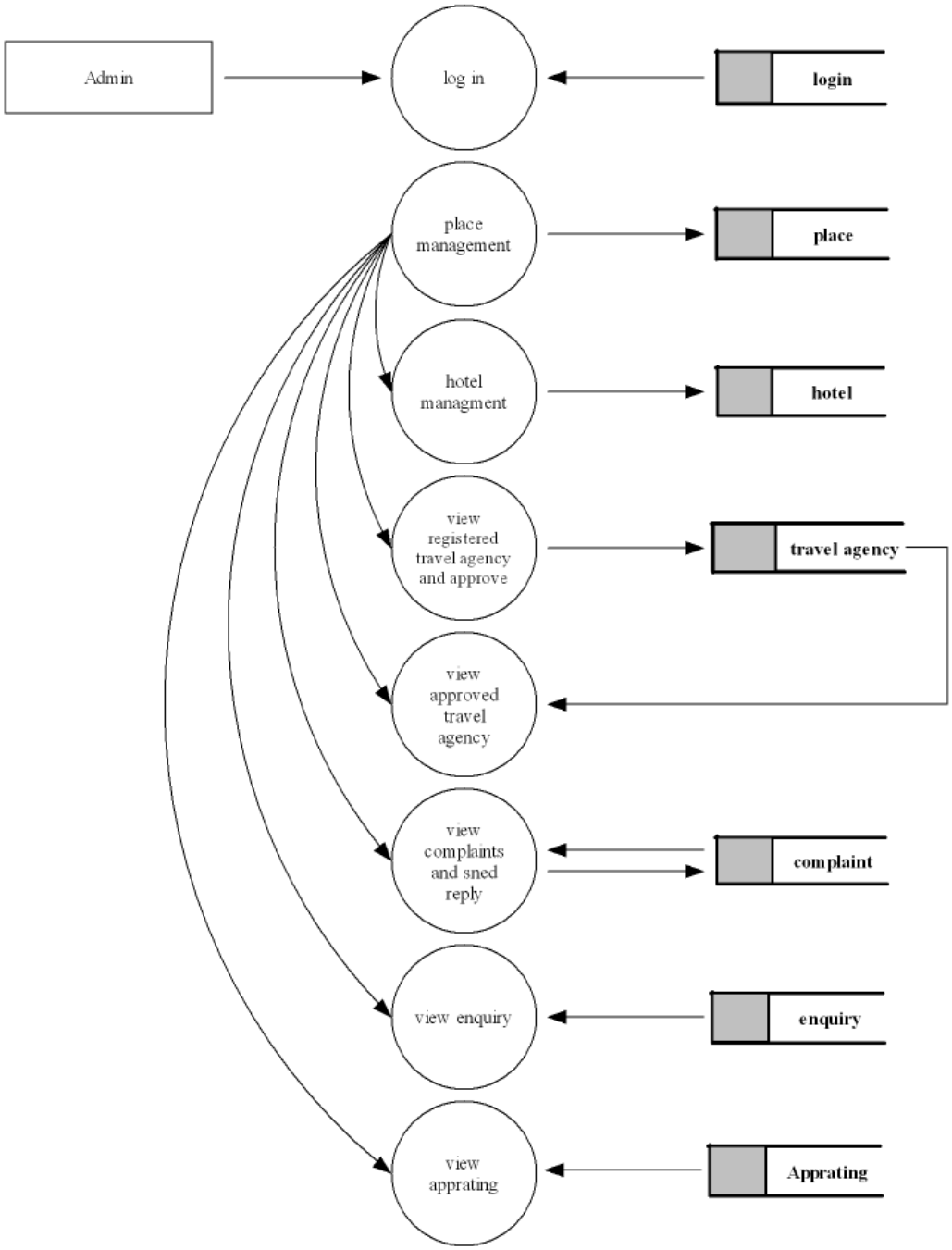
Level 0 DFD



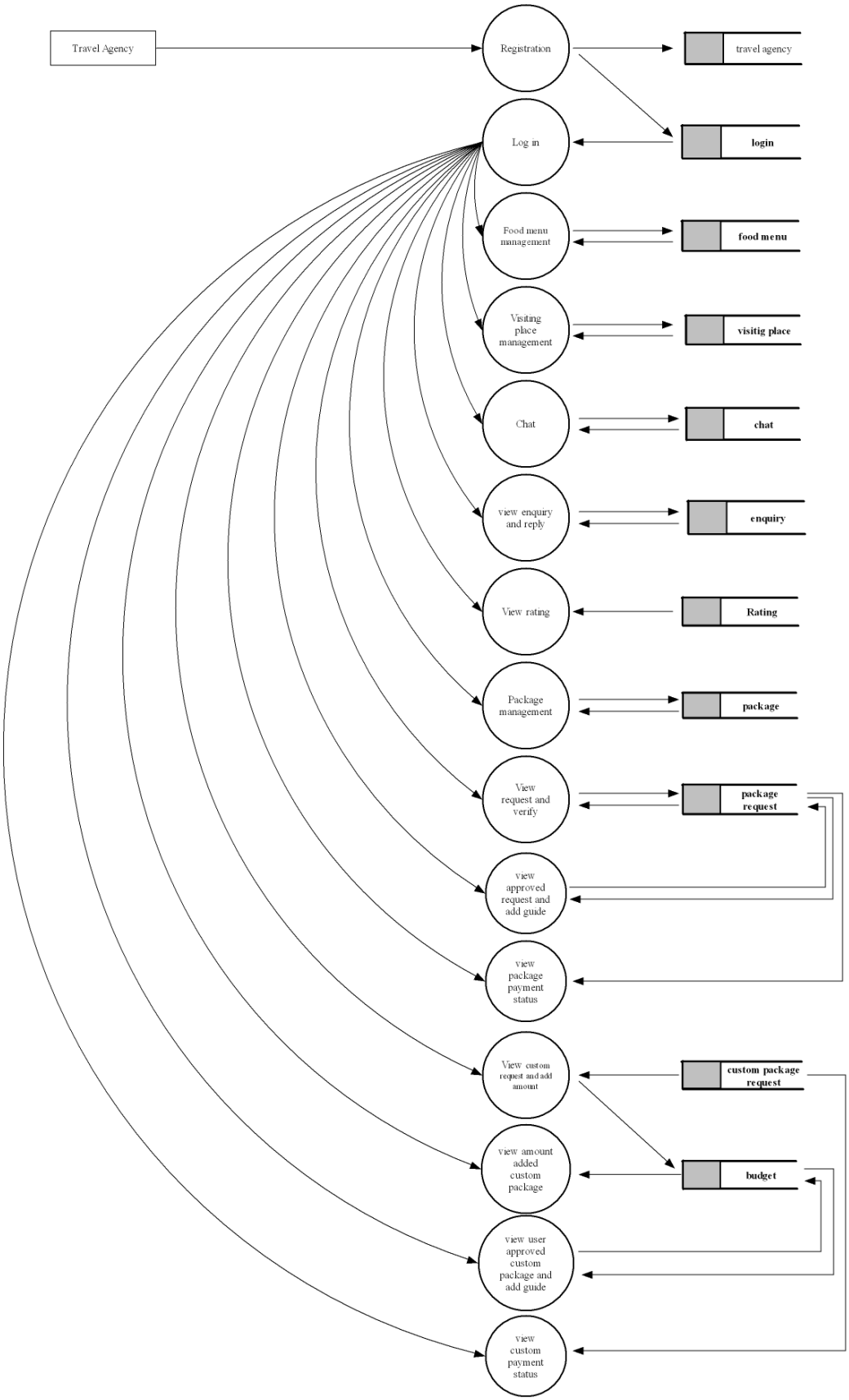
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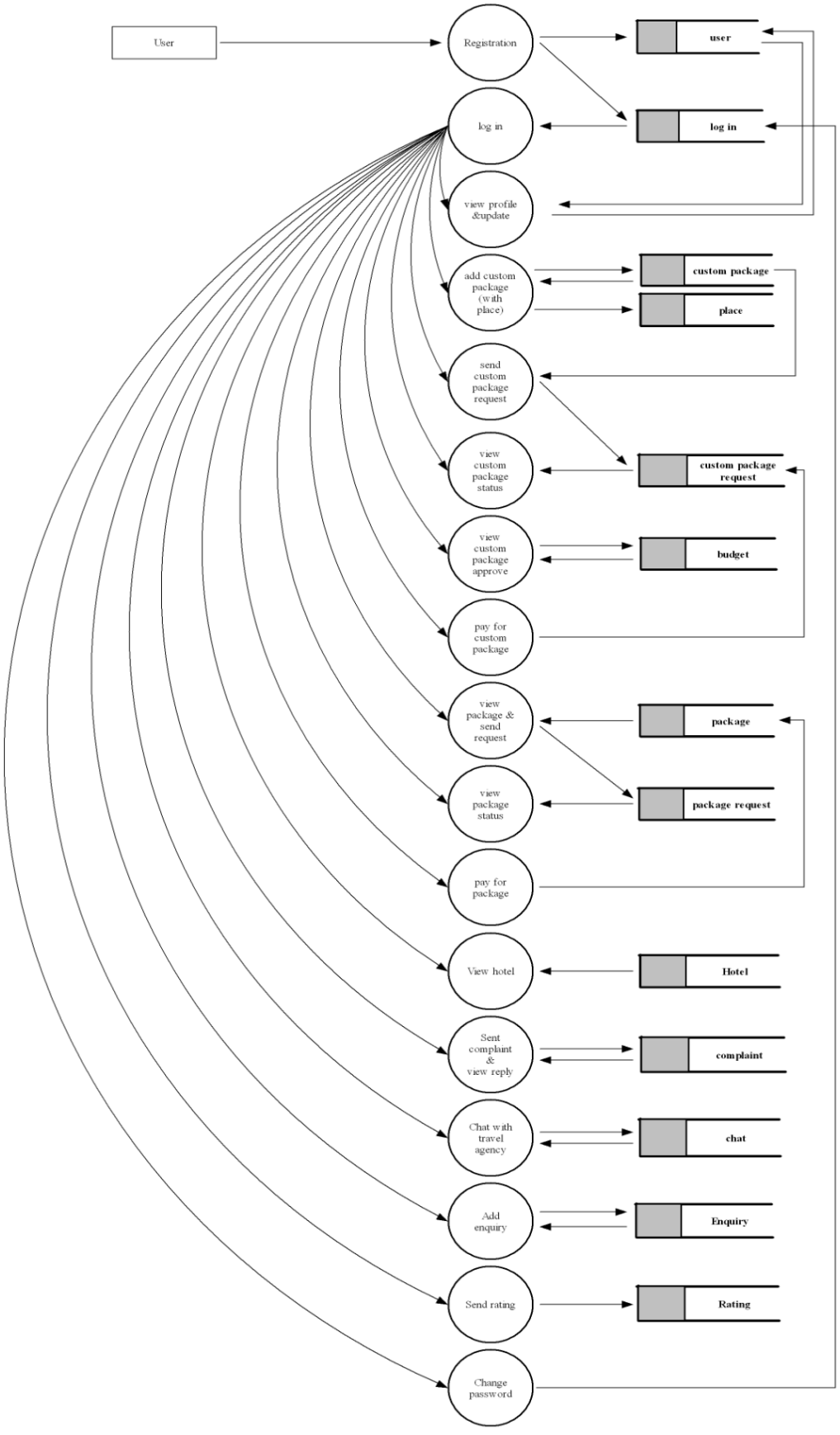
Level 1.1-DFD for ADMIN



Level 1.2-DFD for TRAVEL AGENCY



Level 1.3-DFD for USER



7.SYSTEM TESTING

Testing is the process of executing a program with the intent of finding any errors. Testing is vital to the success of the system. Without proper testing, hide errors will surface after sometime of use and perhaps irreversible damage has been done to valuable data. A series of tests like responsiveness, its value, stress and security are performed before the system Is ready for user acceptance testing. System testing follows the logical conclusion that is all the part of the system are tested and found to be working properly under all kinds of situation s, and then the system is achieving its goal of processing the data perfectly according to the user rules and requirements.

System testing is aimed at ensuring the system works accurately before live operation commences. Testing is vital to the system. System testing makes a logical assumption that if all parts of the system are correct, the goal will be successfully achieved. The candidate system is subjected to a variety of tests: Online response, Volume, Stress Recovery & Security and Usable tests. A series of testing are performed for the proposed system before the system is ready for user acceptance testing. Nothing is complete without, as it is vital success of the system.

7.1 UNIT TESTING

Unit testing focuses verification effort on the smallest unit of the software designs the module. To check whether each module in the software works properly so that it gives desired outputs to the given inputs. All the validations and conditions are tested in the module level the unit test. Boundary condition is tested that the modules operate at boundaries. All independent paths through the control structure ensure that all statements in a module have been executed at least once.

7.2 INTEGRATION TESTING

The major concerns of integration testing are developing incremental strategies that will limit the complexity of entire actions among components as they are added to the system.

Developing a component as they are added to the system, developing an implementation & integration schedules that will make the modules available when needed, designing test cases that will demonstrate the viability of evolving system. Though each program works individually, they should work after linking them together. This is also referred to as Interfacing. Data may be lost across interface and one module can have an adverse effect on another. Subroutines, after linking, may not be do the desired function expected by the main routine. Integration testing is a systematic technique for constructing program structure while at the same time, conducting test to uncover errors associated with the interface. In the testing, the programs are constructed and tested in small segments.

7.3 SYSTEM TESTING

When a system is developed, it is hoped that it performed properly. In practice, however, some errors always occur. The main purpose of testing an information system is to find the errors and correct them. A successful test is one, which finds an error.

The main objectives of system testing are

- To ensure during operation the system will perform as per specification.
- To make sure that the system meets user's requirement during operation.
- To verify that the controls incorporated in the system function as intended.
- To see that when correct inputs are fed to the system the outputs are correct.
- To make sure that during operation, incorrect input and output will be deleted.

The scope of a system test should include both manual operations computerized. Operations system testing is a comprehensive evaluation of the programs, manual procedures, computer operations and controls. System testing is the process of checking if the developed system is working according to the original objectives and requirements. All testing needs to be conducted in accordance tot the test conditions specified earlier.

7.4 DATA VALIDATION TESTING

Data validation testing is done to see whether the corresponding entries made in the tables are correct. Proper validations are done in case of insertion and updating of the tables. If any such case arises, then proper error messages or warning, if any, has to be displayed.

The different test cases are: Guarantee that all independent parts within a module have been exercised at least once.

- Exercise all logical decision on their true/false side.
- Exercise all loops at their boundaries and within their operational bounds.
- Exercise internal data structure to ensure their validity.
- Each module was tested and the tested modules were linked and integration test was carried out.

8.SYSTEM IMPLEMENTATION

Implementation is a stage where theoretical design is turned to the working system. The implementation phase is used to test the developed package with sample data, correcting the error identified, appearing the user of the various special facilities and features of the computerized system. It also involves user training for minimizing resistance to change and giving the new system a change to prove its worth. The successful implementation of the new system depends upon the involvement of the user. Implementation means converting a new or revised system design into an operational one. The implementation includes all those activities that take place to convert from the old system to the new one. There are several activities involved while implementing a project:

- Careful planning.
- Investigating the current system and its constraints on implementation.
- Design of methods to achieve the change over.
- Training of the staff in the change over procedure and evaluation of change over method.

Implementation is the final stage and it is an importance phase. The first task in implementation was the implementation planning, that is deciding on methods to be adopted. After the system was implemented successfully, training of the user was one of the most important subtasks of the developer. For this purpose the user or system manual were prepared and handled over to the user to operate the developed system. So change over plays a vital role, which checks the developed tool for the following requirements, and then only the user accepted the developed tool. The change over took place only when the system had been proved to the satisfaction of the system analysis and other implementation activities have been completed.

8.1 IMPLEMENTATION METHODS

There are several methods for handling the implementation and consists for changing from the old to the new computerized system. The most secure method for conversion from the old system is to run the old and new system in parallel .In this approach; a person may operate in the manual processing system as well as start operating the new computerized system.

Another commonly used method is a direct cut over the existing manual system to the computerized system. The change may be with in a week or a day. This strategy requires planning. A working version of the system can also be implemented in one part of the organization and the changes can be made as and when required, but this method is less preference due to the loss of entire system. After the system is implemented, a review should be conducted to determine whether the system is meeting expectations where improvements are needed.

8.2 IMPLEMENTATION PLAN

Implementation plan includes a description of all activities that must occur to implement the new system and to put into operation. It defines the personnel responsible for the activities and prepares a time chart for Implementation of the system. The Implementation plan should anticipate possible problems and must be able to deal with them. The usual problem may be missing documents, missed data formats between current and new files, errors in data translation, missing data etc.

8.3 USER TRAINING

After the system is implemented successfully, training of the user is one of the most important subtasks of the developer. Even well designed and technically elegant systems can succeed or fail, because of the way they are operated and used. For this purpose user manuals are prepared and handled over to the user to operate the

Developed system. Thus the users are trained to operate the developed system. Both the hardware and software securities are made to run the developed systems successfully in future. In order to put new application system into use, the following activities were taken care of:

- Preparation of user and system documentation
- Conducting user training with demo and hands on
- Test run for some period to ensure smooth switching over the system

The users are trained to use the newly developed functions. User manuals describing the procedures for using the functions listed on menu and circulated to all the users. It is confirmed that the system is implemented up to users need and expectations.

9.CONCLUSION

In this project entitled "**TRAVEL AND TOURISM**", first an attempt has been made to find the need for the system. To fulfill the needs, a detailed study was done to find out the various requirements of the system and it has been designed with an easy to use interface, so that even a user with minimum knowledge is able to operate the system easily.

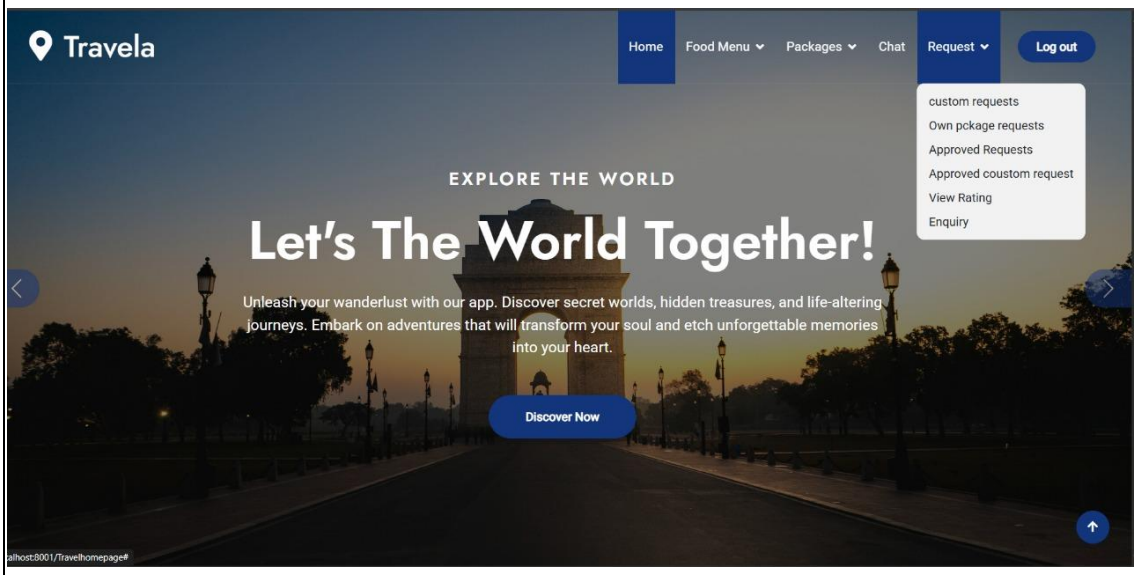
The software is developed with a modular approach. All modules in the system have been tested and everything worked successfully. Thus the new system has fulfilled the objectives identified and is ready to replace the existing system.

The system is well efficient, and in future, any modifications or enhancements if required, can be done to suit the requirements of the time. Also further modules can be added to it to add more features.The website that we developed is working properly. It is very user friendly so that the users can easily interact with. The users are satisfied with the site. We are going to release an enhanced version of our website which is more user friendly than the existing an more error free.
All the requirements specifications was followed as for as possible and few additional features were added that can make the application more user friendly and less complicated.

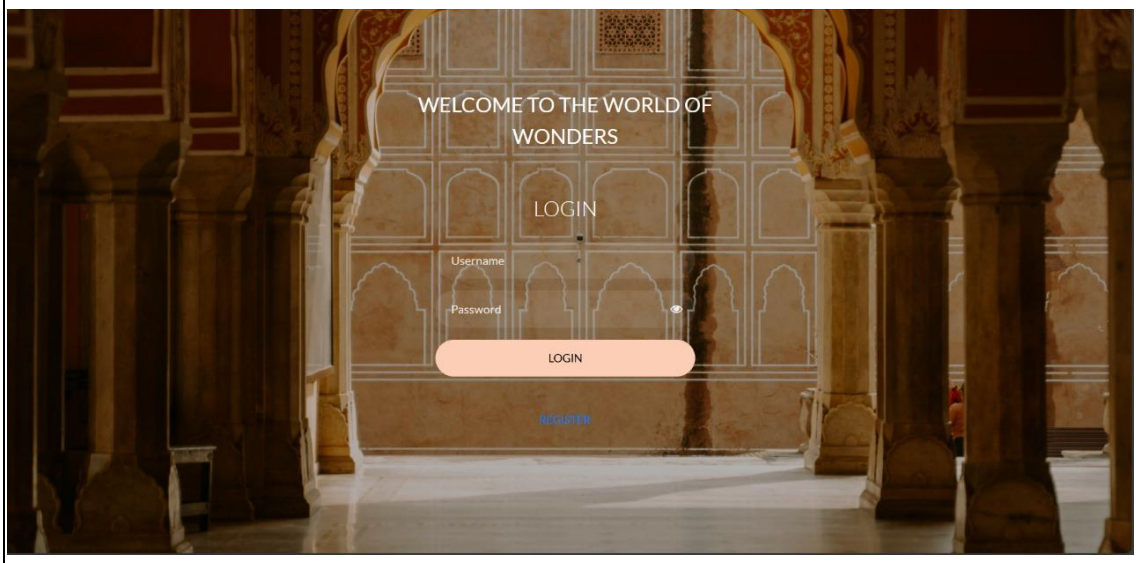
10.APPENDIX

A.1 SCREEN SHOTS

HOME

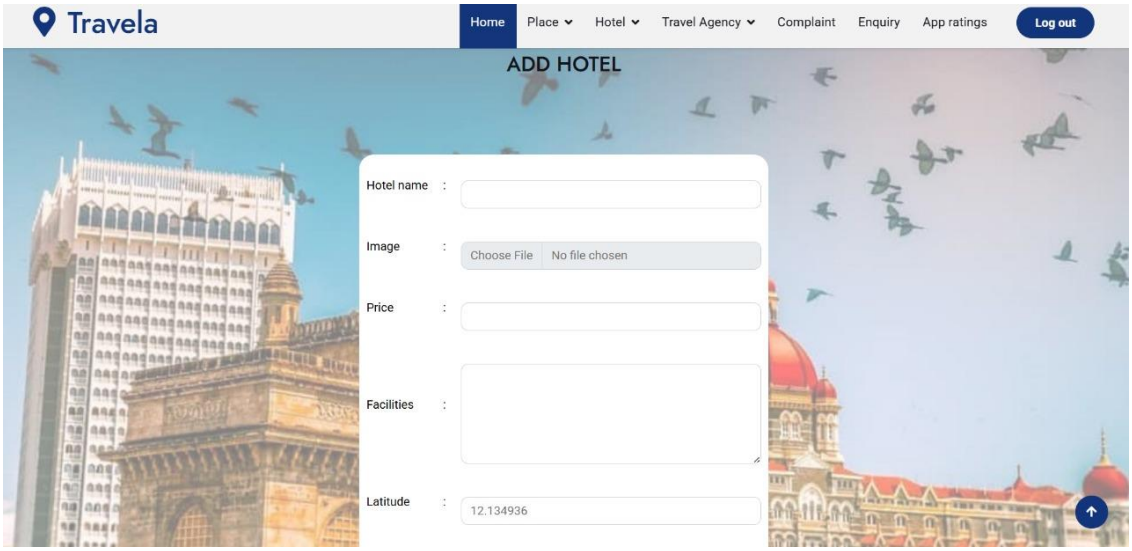


LOGIN

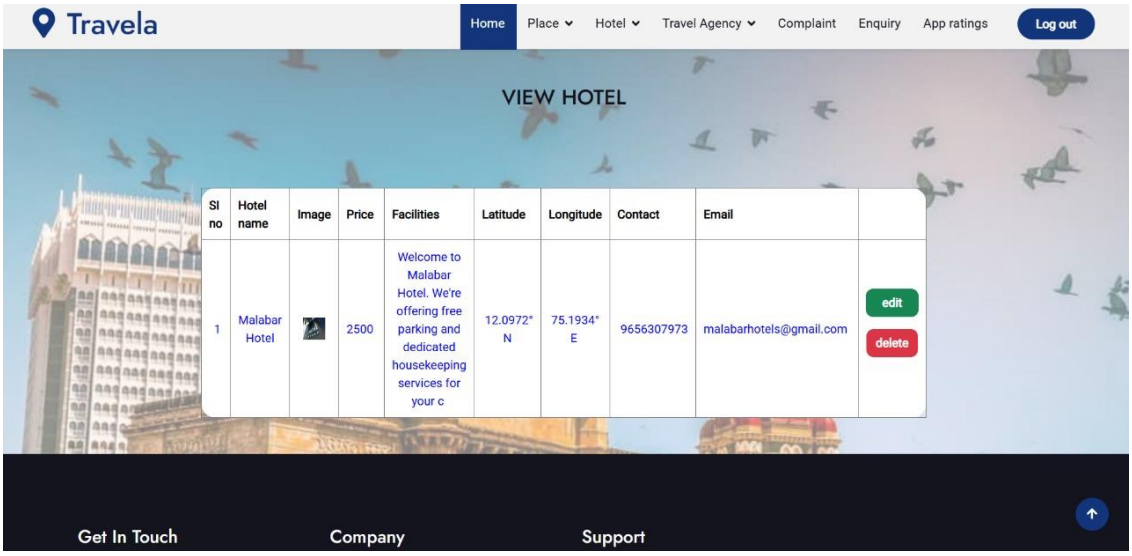


➢ ADMIN

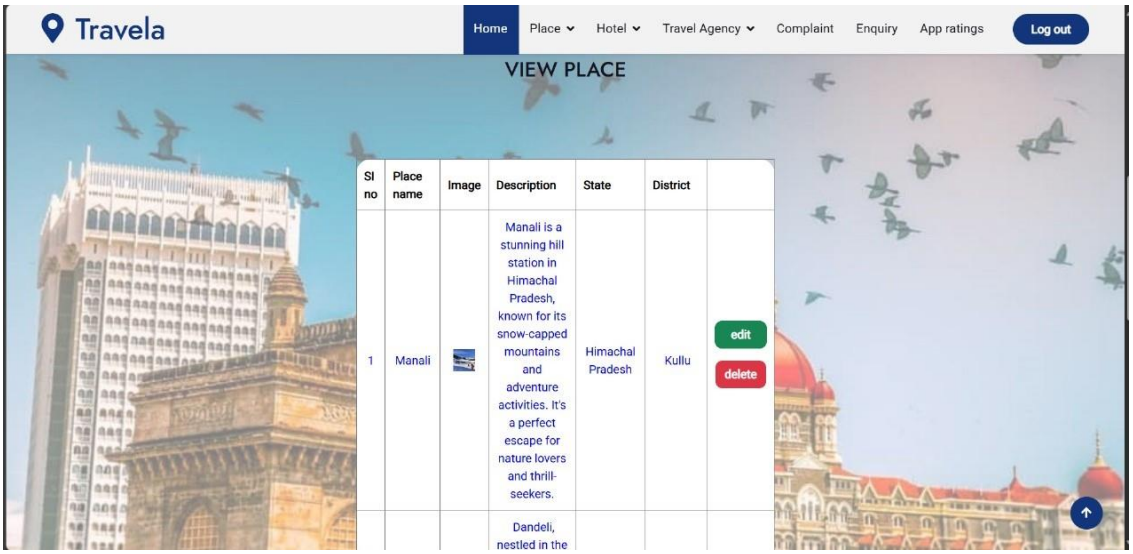
ADD HOTEL



VIEW HOTEL

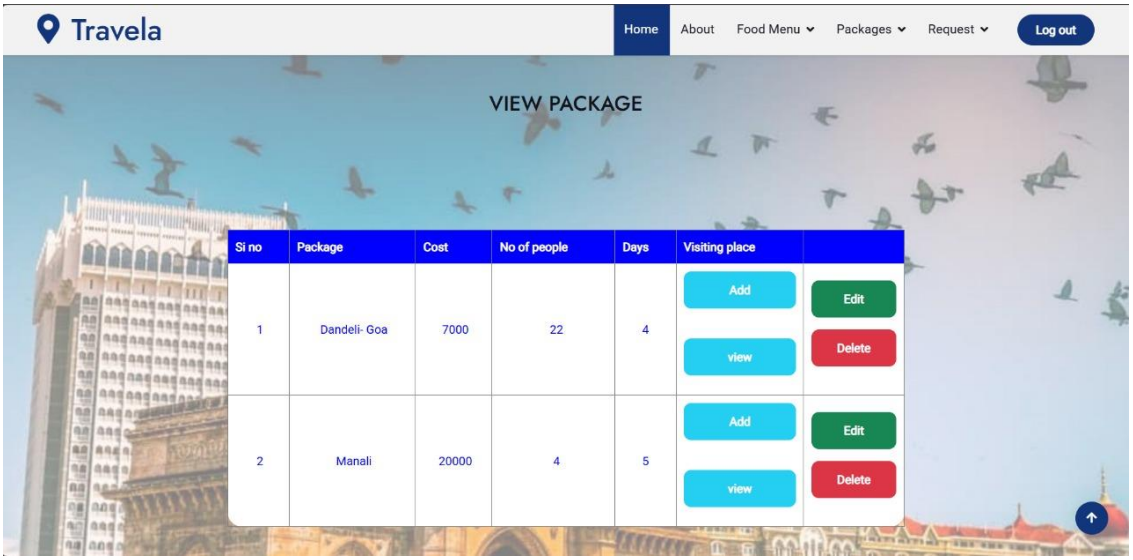


VIEW PLACE

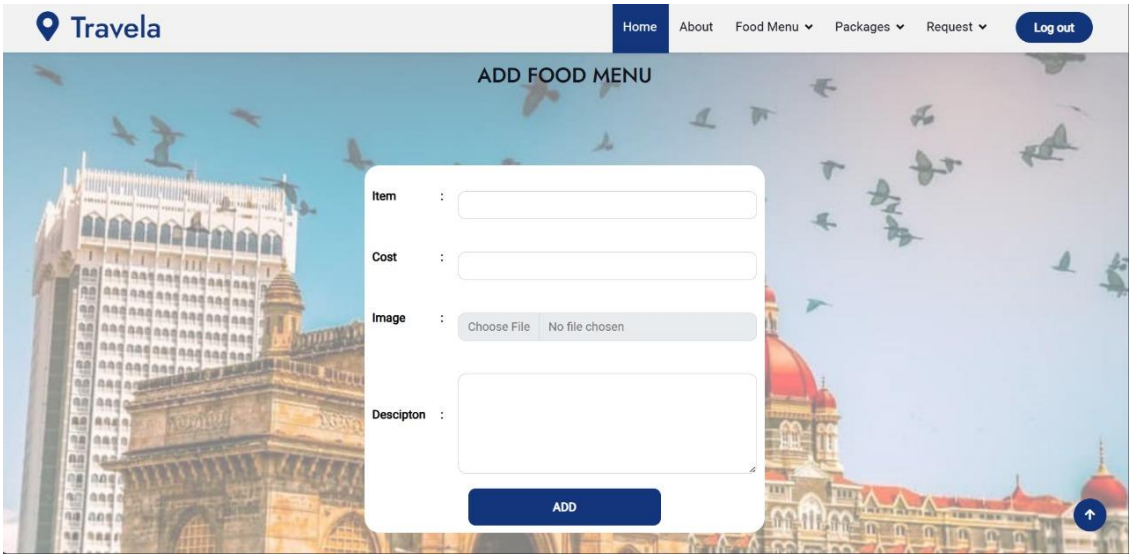


➢ TRAVEL AGENCY

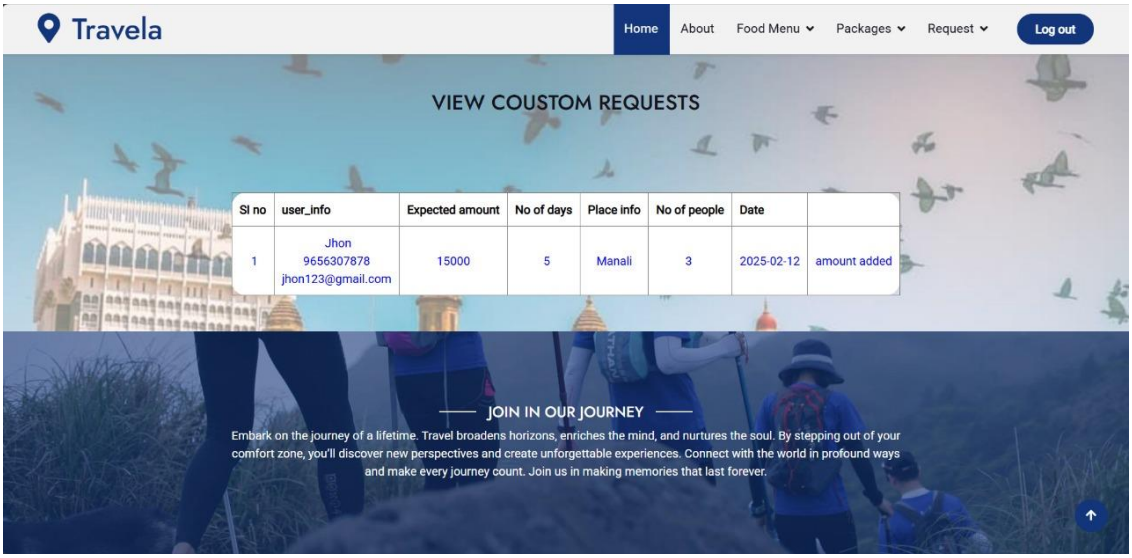
VIEW PACKAGE



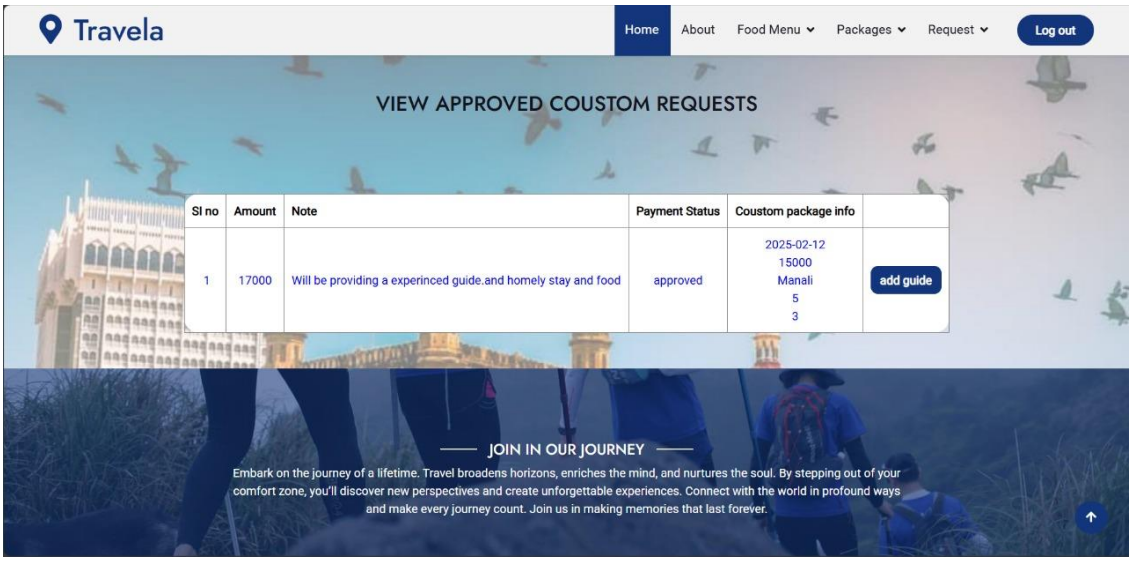
ADD FOOD MENU



VIEW CUSTOM REQUEST



VIEW APPROVED CUSTOM REQUEST

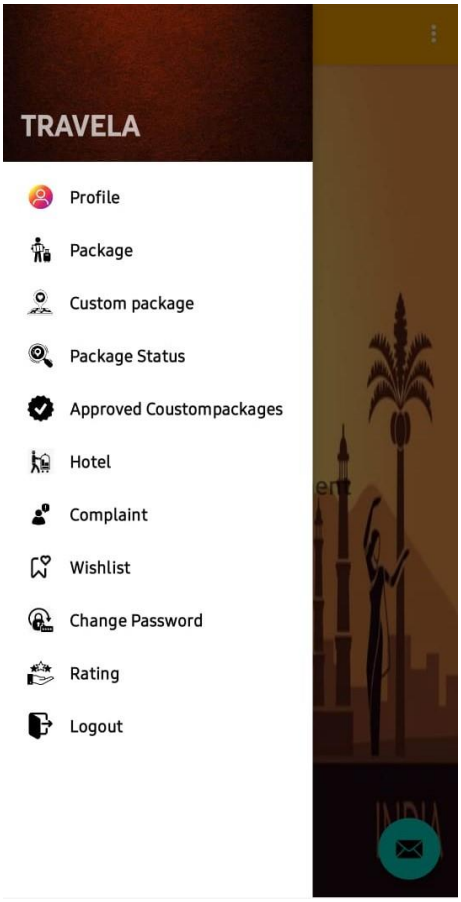


➢ USER

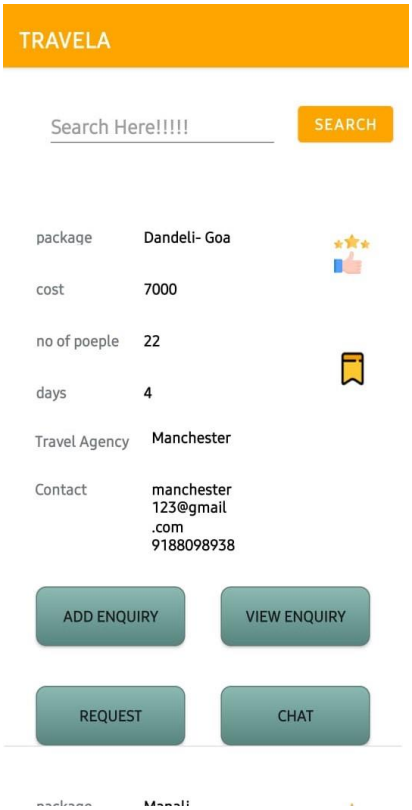
LOGIN



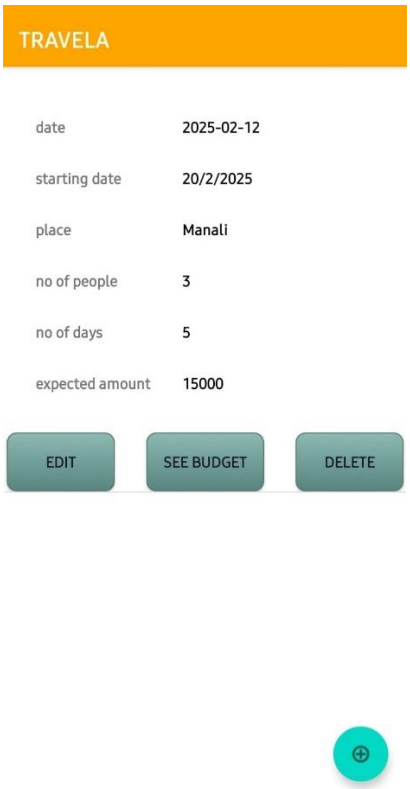
USER HOME



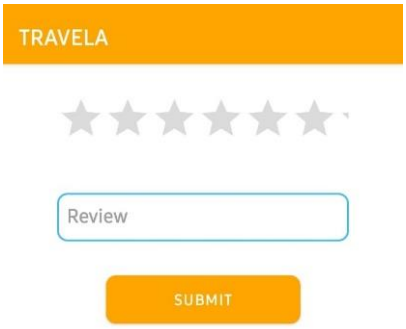
VIEW PACKAGE



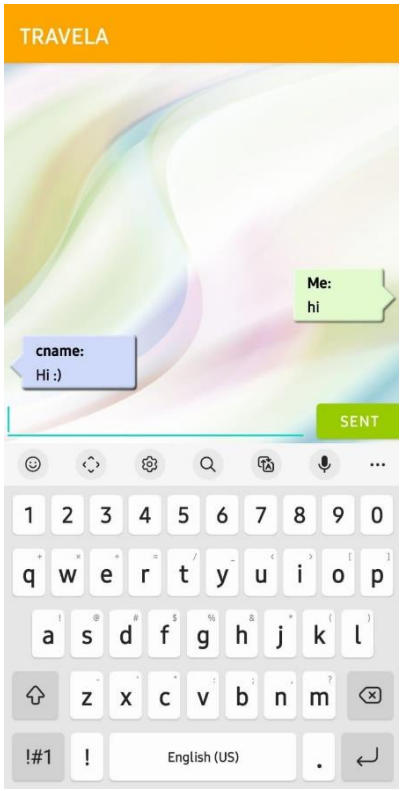
VIEW CUSTOM PACKAGE



RATING



CHATBOT



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2. **Software Engineering a practioner's approach by "R.S Pressman".**
3. **"Abraham Silberschatz, H F Koth, S Sudershan" Database System Concepts 5th Edition.**
4. **Introduction to Computer Science and Programming Using Python Provided by Massachusetts Institute of Technology (MITx).**

Website:

www.w3school.com

www.developphp.com

https://www.python.org/